

PIXELFORGE

Conditional GAN for Pixel Art Sprite Generation

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1. Project Overview

PixelForge is a Conditional WGAN-GP trained from scratch in PyTorch to generate 64×64 pixel art character sprites conditioned on view angle (front, back, left, right). Using the TinyHero dataset (3,648 sprites, Kaggle), the model learns novel, class-conditioned sprite generation and supports smooth latent-space rotation between viewing angles via a Gradio GUI.

2. Model Architecture

Generator

Noise (100-dim) + one-hot label (4-dim) → Linear projection → reshape (512, 4, 4) → 4× ConvTranspose2d (4→64px) → Sigmoid. **Sigmoid over Tanh**: preserves hard pixel art color boundaries. BatchNorm2d + ReLU after each upsampling layer.

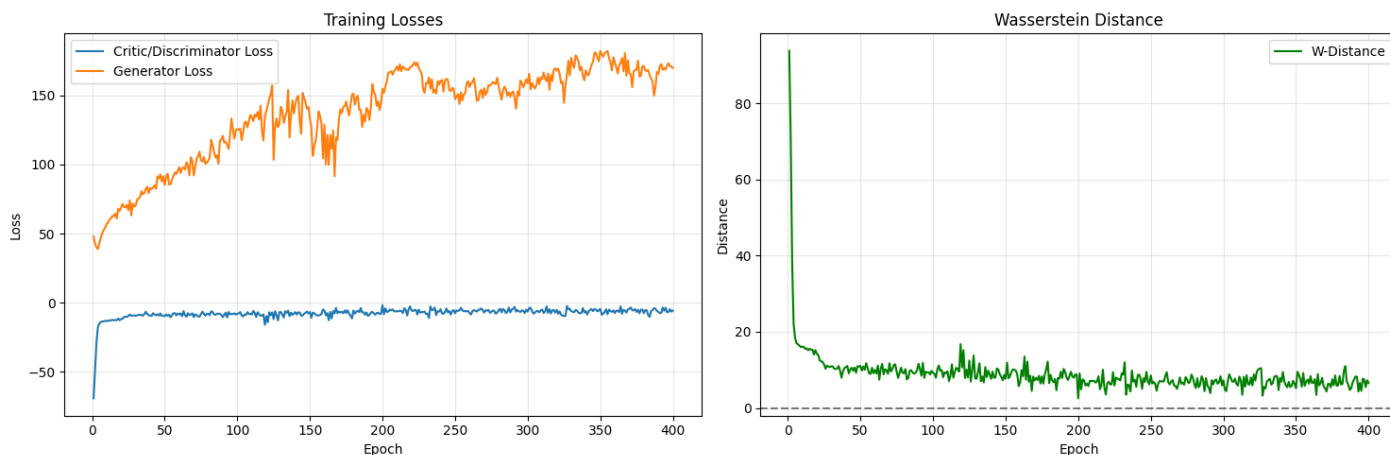
Critic (Discriminator)

Image (3, 64, 64) + class embedding map (1, 64, 64) → 4× strided Conv2d → raw scalar score (no Sigmoid).

InstanceNorm2d over BatchNorm2d: required so the gradient penalty's second-order gradients are not corrupted by inter-sample batch statistics.

WGAN-GP Loss & Key Hyperparameters

WGAN-GP replaces BCE with Wasserstein distance: $L_E = E[D(\text{fake})] - E[D(\text{real})] + 10 \cdot GP$, $L_G = -E[D(\text{fake})]$. Gradient penalty enforces 1-Lipschitz on the critic, providing non-vanishing gradient to G regardless of critic confidence — directly fixing the collapse seen in v1.

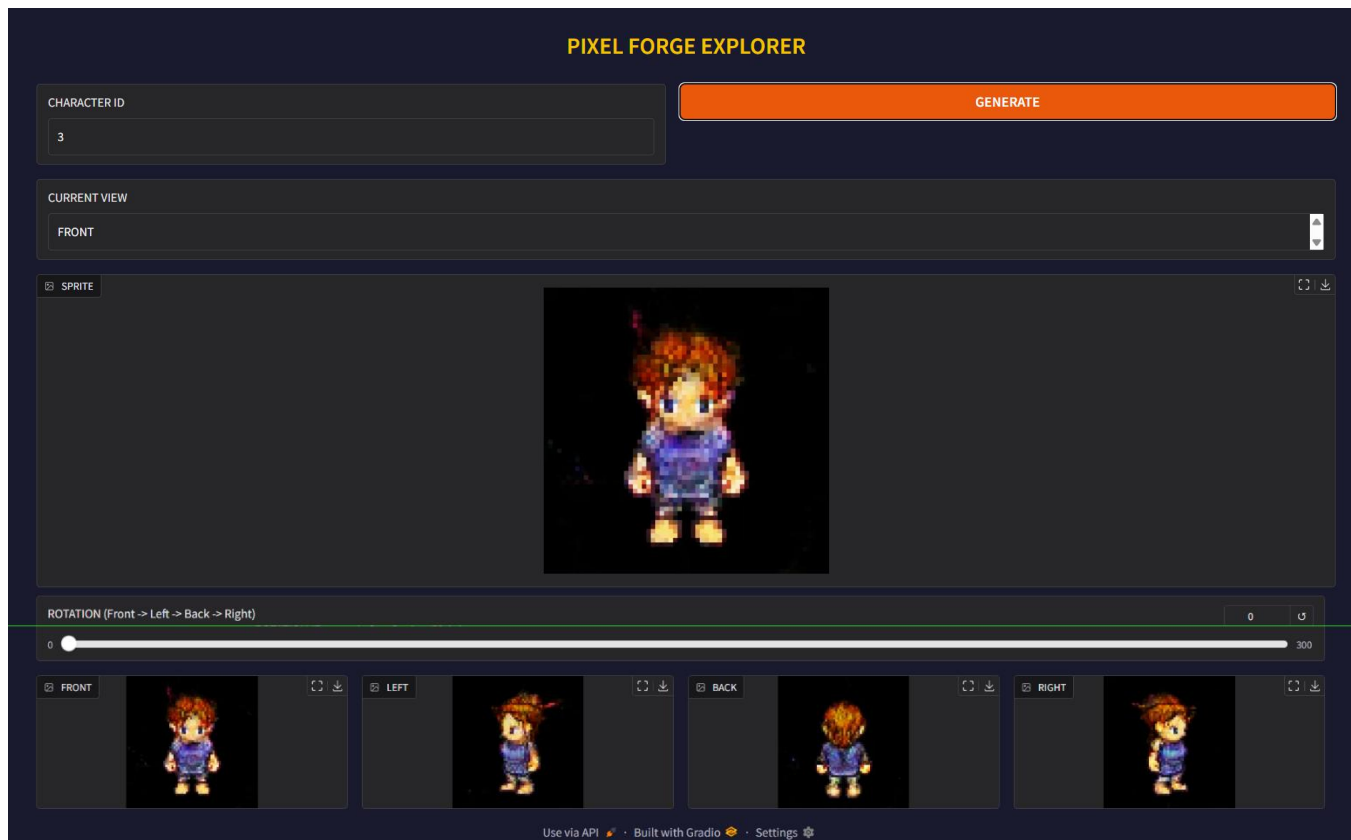


Epochs	Batch	LR	Adam β_1/β_2	D_STEPS	GP_λ	Grad Clip
400	64	1e-4 → 5e-5 (ep.200)	0.0 / 0.9	5	10	max_norm=1.0

3. Extra Criteria

Gradio GUI — Character Angle Explorer

Users enter a character number (0–900) — mapped deterministically to a noise seed — and all four view angles are generated as thumbnails. A rotation slider (0–300, three 100-step segments) SLERP-interpolates conditioning vectors between adjacent angles (front→left→back→right), producing smooth angular transitions while keeping character identity fixed. Sprites are upsampled 4×5× with nearest-neighbour for crisp pixel art. Runs in Colab via `demo.launch(share=True)`.



Hyperparameter Sweep ([hyperparameter_tuning.py](#))

12 configurations across $lr \in \{1e-4, 2e-4\}$, $batch_size \in \{32, 64\}$, $noise_dim \in \{50, 100, 200\}$, each trained for 25 epochs. Scored by a lightweight FID approximation using discriminator conv features rather than InceptionV3, which is inappropriate for 64×64 pixel art (trained on natural photos).

4. Difficulties & Solutions

- **Discriminator collapse (v1):** BCE loss saturated by epoch 50 ($loss_E=0.007$, $loss_G=6-13$). Switched to WGAN-GP — provides non-saturating gradient regardless of critic confidence.
- **Dataset pivot:** Custom sprite-sheet slicer for OpenGameArt produced unusable pixel fragments. Adopted TinyHero (pre-sliced), repurposing conditioning from character class to view angle.
- **AMP incompatibility (v3):** float16 silently violated the Lipschitz constraint, producing neon noise blobs. Removed AMP; full float32 is required for gradient penalty correctness.
- **Training speed:** $D_STEPS=5$ means 5 critic backward passes per generator step. Mitigated with DataLoader pin_memory, num_workers=4, and LR halving at epoch 200.